PANDAS PRACTICAL ASSIGNMENT

Case Study: Online Store Data Cleaning and Analysis

Scenario	You are a data analyst working for an online technology store. Your manager gives you a CSV file containing customer orders. The dataset contains missing values, incorrect data, inconsistent text and duplicated records. Your task is to clean the data, analyze sales performance and generate reports.
Dataset	File: online_store_orders_dirty.csv Size: 36 records and 8 columns Period: October 2026

1. Dataset Description

Column	Description	Example
Order_ID	Order identification	O001
Order_Date	Order date	2026-10-01
Product	Product name	Laptop
Category	Product category	Electronics
Quantity	Number of items	2
Unit_Price	Price per item	850.00
City	Customer city	Yangon
Status	Order status	Completed

2. Known Data Quality Issues

- Missing Quantity and City values.
- Negative Quantity and negative Unit_Price values.
- The text value two appears in the numerical Quantity column.
- Inconsistent city formatting, such as Yangon and yangon .
- Inconsistent order status values, such as Completed and complete.
- A repeated transaction appears with a different Order_ID.

3. Student Tasks

Exercise 1 – Load and Explore Data

1. Read the CSV file using Pandas.
2. Display the first 5 records and the last 10 records.
3. Display the number of rows and columns, all column names, and data types.
4. Generate descriptive statistics using `describe()`.
5. Count missing values in each column.

Exercise 2 – Data Quality Checking

1. Find all missing values and count them by column.
2. Find duplicate records.
3. Display all unique Product names and City names.
4. Find orders where Quantity is less than or equal to zero.
5. Find orders where Unit_Price is less than zero.

6. Display orders whose Status is not Completed.

Exercise 3 – Data Cleaning

1. Create a copy of the original DataFrame. Do not modify the original CSV file.
2. Remove unnecessary spaces from Product and City.
3. Standardize City names (for example, convert text to consistent title case).
4. Standardize order statuses: change complete to Completed; keep other valid statuses unchanged.
5. Convert Quantity into a numeric data type. Convert invalid values into missing values.
6. Handle invalid values: Quantity must be greater than zero and Unit_Price must be greater than zero.
7. Handle missing values using an appropriate strategy. Explain why you selected that strategy.
8. Remove duplicate orders using a suitable rule. Explain why comparing only Order_ID may not be sufficient to detect repeated transactions.

Exercise 4 – Data Analysis

1. Create a new column named Total_Amount using the formula: Total_Amount = Quantity × Unit_Price.
2. Calculate total revenue from valid Completed orders.
3. Identify the product with the highest revenue and the product with the lowest revenue.
4. Calculate the total number of units sold for each product.
5. Calculate total revenue and number of orders by Category. Identify the best-performing category.
6. Calculate total revenue by City. Find the city with the highest revenue.
7. Count the number of orders for each status: Completed, Pending, Cancelled and Shipped.

Exercise 5 – Data Visualization

1. Create a bar chart showing total revenue by product.
2. Create a bar chart showing revenue by city.
3. Create a pie chart showing the distribution of order statuses.

Exercise 6 – Export Results

1. Export the cleaned dataset as online_store_orders_cleaned.csv.
2. Export the sales analysis report as online_store_sales_report.csv.
3. Reopen both exported CSV files and verify the results.

4. Advanced Challenges

1. Find the top 5 orders with the highest Total_Amount.
2. Calculate the average order value (AOV): AOV = Total Revenue / Number of Completed Orders.
3. Find products with total revenue greater than 100.
4. Convert Order_Date into a datetime data type and extract Year, Month and Day.
5. Group sales by month and visualize monthly revenue.

5. Submission Requirements

Create a folder using your name and submit the following files:

```
Student_Name/
  ├── online_store_orders_dirty.csv
  ├── online_store_orders_cleaned.csv
  ├── online_store_sales_report.csv
  └── pandas_online_store_analysis.ipynb
```

- Complete all exercises and include Python code comments.
- Display the results of each task in the Jupyter Notebook.
- Include at least 3 charts.
- Explain important data cleaning decisions.
- Be prepared to explain your code and analysis.

6. Grading Rubric

Criteria	Points
Load and explore data	1.5
Identify data quality problems	1.5
Data cleaning	2.5
Data analysis	2.0
Data visualization	1.5
Export and verification	1.0
TOTAL	10.0

Important: You must understand your code and be prepared to explain your solutions. Do not simply copy and paste code.